

CCTE Topic Defense PPT Format (Chapter 1 & 2)

SLIDE 1 — Title Slide

- Thesis Title
- Proponent
- BS Computer Science
- Date

Example:

"Deep Learning-Based Garbage Classification Using Computer Vision"

SLIDE 2 — Background of the Study

Explain:

- Real-world problem
- Why image/video analysis is needed
- Why computer vision is appropriate
- Current manual method limitations

Example points:

- Manual monitoring is inefficient
- Human error occurs
- No automated detection system
- Computer vision can automate recognition

SLIDE 3 — Problem Statement

General Problem:

- How to automatically detect/classify garbage using computer vision?

Specific Problems:

1. What dataset will be used?
2. What computer vision model will be applied?

3. How accurate is the model?
4. Can the model detect in real-time?

SLIDE 4 — Objectives of the Study

General Objective:

- To develop a computer vision model for garbage

Specific Objectives:

1. To collect image dataset
2. To preprocess image data
3. To train deep learning model
4. To evaluate detection accuracy
5. To compare model performance

SLIDE 5 — Scope and Delimitation

Scope:

- Image-based detection
- Selected dataset
- Selected model (YOLO, CNN, etc.)
- Accuracy evaluation

Delimitation:

- Limited dataset
- No mobile deployment
- Indoor environment only
- Selected object classes only

SLIDE 6 — Significance of the Study

- School
- Industry
- Researchers
- Future developers
- Automation applications

COMPUTER VISION CORE SECTION

SLIDE 7 — Conceptual Framework (Computer Vision Pipeline)

Image Input → Preprocessing → Model Training → Detection → Output

Example:

Camera → CNN → Classification → Result

SLIDE 8 — Dataset Description

- Source of dataset
- Number of images
- Classes/categories
- Image size
- Train/test split

Example:

- 3000 images
- 3 classes
- 80% training
- 20% testing

SLIDE 9 — Image Preprocessing

- Resize images
- Normalize
- Data augmentation
- Labeling
- Annotation (bounding box)

Example:

- Flip
- Rotate
- Zoom
- Brightness adjustment

SLIDE 10 — Computer Vision Model

Choose one:

Traditional ML:

- CNN

Deep Learning:

- YOLOv5
- YOLOv8
- ResNet
- MobileNet
- EfficientNet

Explain why chosen.

SLIDE 11 — Training Process

- Epochs
- Batch size
- Learning rate
- Training environment (GPU/CPU)

Example:

- 50 epochs
- batch size 16
- learning rate 0.001

SLIDE 12 — Evaluation Metrics (VERY IMPORTANT)

For Computer Vision use:

- Accuracy
- Precision
- Recall
- F1 Score
- Confusion Matrix

For Object Detection:

- mAP (mean average precision)
- IoU

SLIDE 13 — Expected Output

- Detected object label
- Bounding box
- Confidence score
- Classification result

Example:

"Fire detected – 98% confidence"

CHAPTER 2

SLIDE 14 — Review of Related Literature

3–5 related computer vision studies

Format:

Author (Year)

Method

Dataset

Accuracy

Relation

SLIDE 15 — Review of Related Computer Vision Systems

System 1 — YOLO-based

System 2 — CNN-based

System 3 — MobileNet-based

Compare them.

SLIDE 16 — Research Gap

Example:

- Low accuracy
- No real-time detection
- Small dataset
- No deep learning model
- No multi-class detection

SLIDE 17 — Proposed Computer Vision Framework

Dataset

- Image Preprocessing
- CNN / YOLO Model
- Training
- Testing
- Detection Output

SLIDE 18 — Expected Contribution

- Automated detection
- High accuracy model
- Real-time detection
- New dataset
- Model comparison

SLIDE 19 — Summary

- Problem
- Dataset
- Model
- Evaluation
- Expected Output

SLIDE 20 — Thank You

Questions?